

Certified Contraction of Reachable Sets under Sequential Telemetry

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Abstract

Keywords: reachable sets; occupation measures; set-membership estimation; data assimilation; Koopman operator; ultraspherical spectral methods; hybrid systems; atmospheric entry

MSC 2020: [select five from the list in the source]

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Open items

everything here is conditional on paper 1: the multi-phase occupation-measure embedding and the derived certificate (Φ, Ψ)	6
make-or-break condition for this paper: per "epoch", does measurement restriction contract faster than wrapping inflates? if wrapping wins there is no contraction theorem (bound the number of "epochs" over which contraction is strict?).	8

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1.1 Static certified bounds and their blind spot

1.2 Set-membership estimation

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2.2 Measurement epochs and the observation model

2.3 What a bounded-error return certifies

2.4 The sequential reachability problem

2.5 What contraction can and cannot mean

3 Measure Restriction and the Conditioned Problem

- 3.1 Restriction of an occupation measure
- 3.2 The conditioned Liouville problem
- 3.3 Piecewise propagation and re-initialization
- 3.4 The wrapping effect
- 3.5 Restriction on the reduced object

4 Contraction

4.1 Monotonicity under support restriction

4.2 One-shot contraction

4.3 Sequential contraction: the competition

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7 Discussion

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References

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